

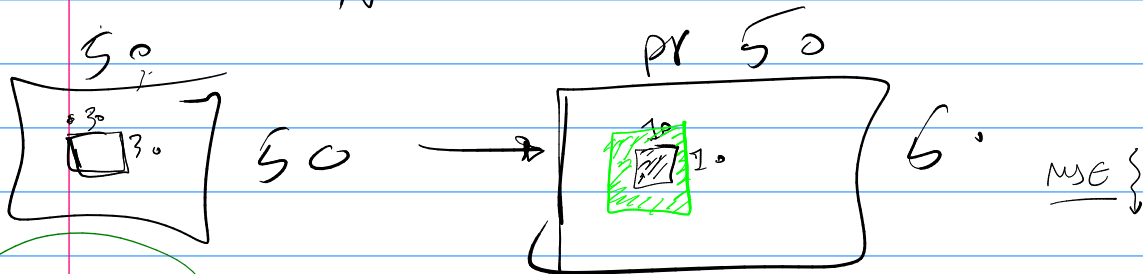
Scale and Shift

Invariant loss!

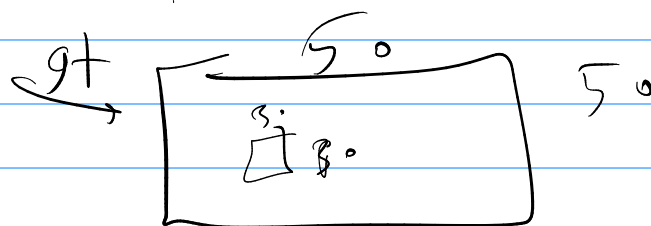
DPT $\rightarrow$ MSE

$(X_i, y_i) \rightarrow d_i$ pr
 $\rightarrow d_i^*$ gt

$$MSE = \frac{1}{N} \sum (d_i - d_i^*)^2$$



✓ scale change!
 ✓ shift



scale and shift inv.

~~MSE~~

$$d = d^* + t$$

$$\min MSE = \frac{1}{N} \sum (d_i + d_i^*)^2$$

$$d_i^* + t$$

Train

$$\text{loss (MSE)} \xrightarrow{s=\sqrt{t}} \frac{1}{N} \sum (d_i - d_i^*)^2$$

scale and shift $\frac{(d) + t}{x}$

inv.

$$\text{MSE} = \frac{1}{N} \sum ($$

pr

$$d^* = \begin{matrix} (0,0) & (0,1) & (0,2) & (0,3) \\ \downarrow \\ 1, & 2, & 4, & 8 \end{matrix}$$

gt

$$d = \begin{matrix} (0,0) & (0,1) & (0,2) & (0,3) \\ \downarrow \\ 11, & 12, & 14, & 18 \end{matrix}$$

MSE = 0

$$\left. \begin{matrix} s = \frac{1}{2} \\ t = 10 \end{matrix} \right\}$$

$$d = \frac{1 \cdot d_{pr} + 10}{10 + x}$$

pr+new

gt

$$s = \frac{1}{2}$$

$$t = 10$$

$$\underline{d^*} = (1, 2, 4, 8)$$

$$\underline{d_P} = (11, 12, 14, 18)$$

$$\text{JSE} = \sqrt{\frac{1}{4} \sum (1-11)^2 + (2-12)^2 + \dots}$$

$$s=1, t=1$$

$$d = s \underline{d^*} + t \underline{10}$$

$$\text{JSE} : \frac{1}{4} \sum (11-11)^2 + (12-12)^2 \dots$$

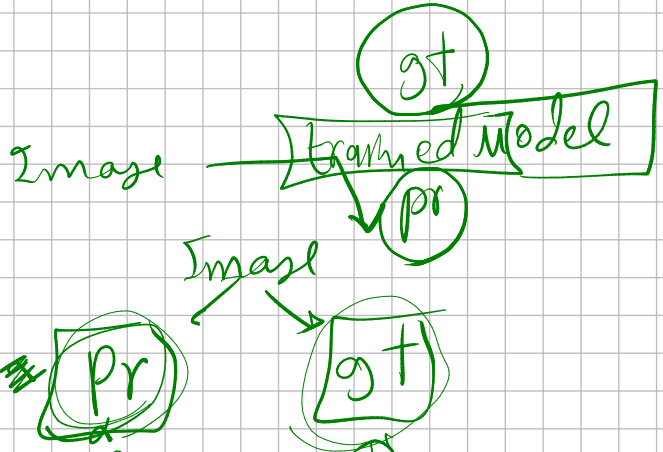
Scale on
x-axis

$$\text{JSE} = 0$$

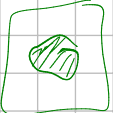
DS1
DS2
DS3

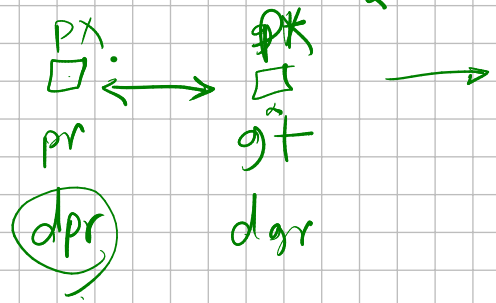
~~Train!~~

MSE



$$\text{MSE} / S / S / \text{Min} = \text{Min} \quad \frac{1}{N} \sum (S d^* + t - d^*)^2$$

pr =  , gt



real sn -

$$\text{MSE} = \text{Min} \quad \frac{1}{N} \sum (S d^* + t - d^*)^2$$

s, t
↓ min

s, t
α

✓

ms t

s, t
α

2ms
3. 3.
—

Algorithm
Dpt

gt
3. 3.
9

✓
10
9
10
PO

$\sim$ KZTT I $\rightarrow$ DPT $\rightarrow$ $\begin{matrix} (0,0) \\ \text{gt} \end{matrix} \rightarrow d = 10m$ $\rightarrow$ MSE $\checkmark$
 $\text{pr} \begin{matrix} (0,0) \\ \rightarrow \end{matrix} d = 10m$

$\checkmark$ $\leftarrow$ α
 $\sim$ KZTT I $\rightarrow \checkmark$
 $\checkmark \leftarrow$ NYU $\rightarrow \checkmark$
 $\checkmark$ DPT $\rightarrow$ $\begin{matrix} \alpha \\ \text{DPT} \\ \alpha \end{matrix}$
 $\checkmark$ D53
 $\begin{matrix} (0,0) \\ \text{gt} \end{matrix} \rightarrow 10m$
 $\begin{matrix} (0,0) \\ \text{gt} \end{matrix} \rightarrow 8m$
 $\begin{matrix} +2 \\ \text{Sd} + t_x \end{matrix}$
 $MSE = \frac{1(8-10)^2}{1} = 4$

Brimmed Qass

دستور کمر کشف → امار و صفا

مجموعه

1.7%

MSB

میرک